snowflake

TAKE-HOME ASSIGNMENT

To run SQL queries as cheaply as possible, Snowflake's query optimizer rewrites them into low-level plans that are more efficient but return the same results. For this exercise, we've made MiniVM, a simplified version of the engine that runs those plans. A MiniVM program is a list of instructions acting on columns of values. Below is an example.

MiniVM program

```
LOAD    a, orders.o_amt
GT      m, a, 100.00
IDX     i, m
GATHER  big, a, i
OUT     big
```

Equivalent SQL query

```
SELECT o_amt
FROM   orders
WHERE  o_amt > 100
```

TASK

Customers have many legacy MiniVM programs that run slowly. Your job is to ship an **optimizer** that rewrites any program so it gives the same results in fewer cycles.

A bad rewrite could silently change what a program does, so you also need to build an **evaluator** that checks whether an optimizer is correct.

REQUIREMENTS

`./optimize program.mini schema.json` prints an optimized program. `schema.json` has the tables, constraints, and stats of the data it will run on.

- Same results on any data with the tables and constraints in `schema.json`
- Order doesn't matter unless the input program specifies it
- Numbers must agree within 1e-6 relative
- If the input program errors, anything goes
- At most 1 second per program

`./evaluate OPTIMIZER` tests any program with the same interface as `optimize`. Exits 0 if it finds no defects, non-zero if it finds any.

- A defect is an optimized program whose results differ, by the rules on the left
- 2 minutes at most
- It may use `bin/minivm`

WHAT TO SEND

1. Everything behind your optimizer and evaluator: code, scripts, prompts, docs, logs, your agents' raw session files (not exports or summaries), and your git history (commit as you go). We look at it all to follow how you worked. *Do not send us any binary files*, including `bin/minivm`.
2. A **written plan**: what would you do with a few days and unlimited compute and tokens?
3. Which AI models you used, and roughly how many tokens or how much spend.

We will judge the whole submission, not just the final programs. Your optimizer should work on programs and data we haven't seen, and make them noticeably faster.

Expect to spend 2–3 hours of your own time. Use of AI is necessary, and you may use any tools you like. Start with `docs/README.md` in the attached zip, and ask us for a build if `bin/minivm` doesn't run.